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Last login: Sun Jun 28 06:46:13 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % echo "==== MiniMD folder layout ====="
ls -la "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD"
echo ""
echo "==== subfolders (one level down) ====="
for d in "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD"/*;
do echo "---- $d ----"; ls "$d" | grep -E "\.py$|\.sh$|results"; done
==== MiniMD folder layout =====
total 216
drwxr-xr-x   9 stephenrajva  staff   288 24 Jun 23:24 .
drwxr-xr-x@  8 stephenrajva  staff   256 24 Jun 23:39 ..
-rw-r--r--@  1 stephenrajva  staff 10244 28 Jun 06:07 .DS_Store
drwxr-xr-x  10 stephenrajva  staff   320 19 Jun 13:03 Comparison
drwxr-xr-x  12 stephenrajva  staff   384 24 Jun 23:19 DragonFly
drwxr-xr-x  12 stephenrajva  staff   384 19 Jun 12:47 Mesh
-rw-r--r--@  1 stephenrajva  staff 59728 19 Jun 13:05 MiniMD_terminal.pdf
-rw-r--r--@  1 stephenrajva  staff 33629 24 Jun 23:24 Terminal_Overall.pdf
drwxr-xr-x  24 stephenrajva  staff   768 24 Jun 20:31 Torus

==== subfolders (one level down) =====
---- /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Comparison
/ ----
combine_minimd_results.py
plot_minimd_comparison.py
---- /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/
----
analyze_universal.py
minimd_dfly_ppn1.py
results
run_minimd_dfly_sweep.sh
---- /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Mesh/ ----
extract_mesh_metrics.py
minimd_mesh16.py
minimd_mesh64.py
plot_mesh_scaling.py
---- /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/ ---
-
analyze_universal.py
extract_torus_metrics.py
minimd_torus16.py
minimd_torus3d_ppn1.py
minimd_torus64.py
plot_torus_scaling.py
results
run_minimd_torus_sweep.sh
stephenrajva@Stephens-MacBook-Air-4 ~ % find "/Users/stephenrajva/Downloads/HPC-
Upload 2" -name "minimd_mesh3d_ppn1.py" 2>/dev/null
find "/Users/stephenrajva/Downloads/HPC-Upload 2" -name "minimd_mesh*ppn1*.py" 2
>/dev/null
ls "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D
/" 2>/dev/null | grep -E "\.py$"
/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/min
imd_mesh3d_ppn1.py
/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/min
imd_mesh3d_ppn1.py
analyze_minimd_mesh3d.py
hopcount.py
mesh3D.py

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minimd_mesh3d_ppn1.py
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Up
load 2/SST_Stimulator/MiniMD/DragonFly"
mkdir -p PPN_2
mv ~/Downloads/minimd_dfly_ppn2.py ~/Downloads/run_minimd_dfly_sweep_ppn2.sh PPN
_2/ 2>/dev/null
ls -l PPN_2/
total 0
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd "/Users/stephenrajva/Downloa
d s/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/PPN_2"
chmod +x run_minimd_dfly_sweep_ppn2.sh
export DFLY_SIZE="4"; export STATS_DIR="./results"
sst minimd_dfly_ppn2.py
chmod: run_minimd_dfly_sweep_ppn2.sh: No such file or directory
Failed to canonicalize path [minimd_dfly_ppn2.py]: No such file or directory
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % find ~/Downloads -name "minimd_dfly_
ppn2.py" 2>/dev/null
find ~/Downloads -name "minimd_torus3d_ppn2.py" 2>/dev/null
ls -lt ~/Downloads | head
/Users/stephenrajva/Downloads/minimd_dfly_ppn2.py
/Users/stephenrajva/Downloads/minimd_torus3d_ppn2.py
total 4815344
-rw-r--r--@ 1 stephenrajva  staff          417 28 Jun 06:57 run_minimd_dfly_swee
p_ppn2.sh
-rw-r--r--@ 1 stephenrajva  staff        4467 28 Jun 06:57 minimd_dfly_ppn2.py
-rw-r--r--@ 1 stephenrajva  staff          451 28 Jun 06:57 run_minimd_torus_swe
ep_ppn2.sh
-rw-r--r--@ 1 stephenrajva  staff       3827 28 Jun 06:57 minimd_torus3d_ppn2.
py
drwxr-xr-x  7 stephenrajva  staff         224 26 Jun 11:43 Report IITK
-rw-r--r--@ 1 stephenrajva  staff     158155 26 Jun 11:26 minimd_ghost_atom_ex
change.png
-rw-r--r--@ 1 stephenrajva  staff     144976 26 Jun 11:26 minife_collective_co
mmunication.png
-rw-r--r--@ 1 stephenrajva  staff     311300 25 Jun 14:59 JD_Digital_Process_E
nablement_Associate 1.pdf
-rw-r--r--@ 1 stephenrajva  staff    2123087 25 Jun 12:08 Report final.pdf
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % mv ~/Downloads/minimd_dfly_ppn2.py ~
~/Downloads/run_minimd_dfly_sweep_ppn2.sh "/Users/stephenrajva/Downloads/HPC-Uplo
ad 2/SST_Stimulator/MiniMD/DragonFly/PPN_2/"
mv ~/Downloads/minimd_torus3d_ppn2.py ~/Downloads/run_minimd_torus_sweep_ppn2.sh
"/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/PPN_2/"
2>/dev/null || { mkdir -p "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimu
lator/MiniMD/Torus/PPN_2"; mv ~/Downloads/minimd_torus3d_ppn2.py ~/Downloads/run
_minimd_torus_sweep_ppn2.sh "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stim
ulator/MiniMD/Torus/PPN_2/"; }
echo "=== DragonFly/PPN_2 ==="; ls -l "/Users/stephenrajva/Downloads/HPC-Upload
2/SST_Stimulator/MiniMD/DragonFly/PPN_2/"
echo "=== Torus/PPN_2 ==="; ls -l "/Users/stephenrajva/Downloads/HPC-Upload 2/SS
T_Stimulator/MiniMD/Torus/PPN_2/"
=== DragonFly/PPN_2 ===
total 24
-rw-r--r--@ 1 stephenrajva  staff        4467 28 Jun 06:57 minimd_dfly_ppn2.py
-rw-r--r--@ 1 stephenrajva  staff          417 28 Jun 06:57 run_minimd_dfly_sweep_ppn2.
sh
=== Torus/PPN_2 ===
total 16
-rw-r--r--@ 1 stephenrajva  staff       3827 28 Jun 06:57 minimd_torus3d_ppn2.py

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-rw-r--r--@ 1 stephenrajva  staff   451 28 Jun 06:57 run_minimd_torus_sweep_ppn2
.sh
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniMD/DragonFly/PPN_2"
chmod +x run_minimd_dfly_sweep_ppn2.sh
export DFLY_SIZE="4"; export STATS_DIR="./results"
sst minimd_dfly_ppn2.py
=====
Topology           : Dragonfly (size 4) - MiniMD
hosts_per_router   : 2
routers_per_group  : 2
num_groups          : 2
Total routers      : 4
Total nodes        : 8
Total MPI ranks     : 8
PPN (ranks/node)   : 2    <-- should be 2
Stats CSV          : ./results/minimd_dragonfly_4_ppn2_nodes8_stats.csv
=====
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161476956324:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
161476956324:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161479.766 us, loop 325, bufLen 20510, latency 248.430 u
s. bandwidth 0.082558 GB/s
161479946840:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161479946840:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
161737554558:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
161737554558:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
161740411674:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
161740411674:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
317174430808:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
317174430808:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us
. bandwidth 0.037573 GB/s
317175772907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
317175772907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
317521472242:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
317521472242:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
317522947741:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
317522947741:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
478657028438:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
478657028438:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161484.246 us, loop 325, bufLen 20510, latency 248.437 u
s. bandwidth 0.082556 GB/s
478660018954:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
478660018954:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
479264290584:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
479264290584:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
479267147700:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
479267147700:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us
. bandwidth 0.037573 GB/s

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Allreduce: ranks 8, loop 40, 1 double(s), latency 2473.533 us
Barrier: ranks 8, loop 15, latency 1397.265 us
Simulation is complete, simulated time: 754.264 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniMD/DragonFly/PPN_2"
./run_minimd_dfly_sweep_ppn2.sh
ls -l results/*.csv
=== MiniMD dragonfly size 4 (PPN=2) ===
=====
Topology          : Dragonfly (size 4) - MiniMD
hosts_per_router  : 2
routers_per_group : 2
num_groups        : 2
Total routers     : 4
Total nodes       : 8
Total MPI ranks   : 8
PPN (ranks/node)  : 2    <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/DragonFly/PPN_2/results/minimd_dragonfly_4_ppn2_nodes8_stats.csv
=====
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161476956324:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
161476956324:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161479.766 us, loop 325, bufLen 20510, latency 248.430 u
s. bandwidth 0.082558 GB/s
161479946840:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161479946840:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
161737554558:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
161737554558:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
161740411674:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
161740411674:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
317174430808:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
317174430808:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us
. bandwidth 0.037573 GB/s
317175772907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
317175772907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
317521472242:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
317521472242:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
317522947741:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
317522947741:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
478657028438:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
478657028438:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161484.246 us, loop 325, bufLen 20510, latency 248.437 u
s. bandwidth 0.082556 GB/s
478660018954:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
478660018954:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
479264290584:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
479264290584:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
479267147700:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
479267147700:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8

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AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us
. bandwidth 0.037573 GB/s
Allreduce: ranks 8, loop 40, 1 double(s), latency 2473.533 us
Barrier: ranks 8, loop 15, latency 1397.265 us
Simulation is complete, simulated time: 754.264 ms
=== MiniMD dragonfly size 16 (PPN=2) ===
=====
Topology          : Dragonfly (size 16) - MiniMD
hosts_per_router  : 2
routers_per_group : 4
num_groups        : 4
Total routers     : 16
Total nodes       : 32
Total MPI ranks   : 32
PPN (ranks/node)  : 2    <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/DragonFly/PPN_2/results/minimd_dragonfly_16_ppn2_nodes32_stats.csv
=====
180000:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
180000:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
180000:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
180000:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
180000:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
180000:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
180000:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
180000:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
180000:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
180000:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
180000:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
180000:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
180000:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
180000:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
180000:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
180000:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
161501205109:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
AllPingPong: total time 161504.015 us, loop 325, bufLen 20510, latency 248.468 u
s. bandwidth 0.082546 GB/s
161504195625:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
161611185405:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
161614109221:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
161615102029:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
161618025845:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32

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161758901893:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
161761759009:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
161779138006:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
161781994839:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
161791723629:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
161794580745:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
161972440609:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
161972440609:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
161973023567:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
161973023567:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
161975364243:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
161975364243:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
161975946945:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
161975946945:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
162096517283:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
162099374345:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
162099605493:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
162102462609:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
162468761312:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
162468761312:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
162469447655:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
162469447655:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
162471618374:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
162471618374:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
162472304872:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
162472304872:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
317198679593:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us
. bandwidth 0.037573 GB/s
317200021692:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
317351881489:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
317353290288:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
317355798113:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
317357206912:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
317542819577:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
317544295076:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
317563055023:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
317564530522:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
317575641313:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
317577116812:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
317713989119:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
317713989119:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
317715359804:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
317715359804:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
317715397918:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
317715397918:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
317716768603:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
317716768603:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
317880434967:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
317881910466:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
317883523177:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
317884998676:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
318252678996:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
318252678996:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
318253433373:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
318253433373:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
318254154495:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
318254154495:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32

```

318254908872:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
318254908872:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
478677864184:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
AllPingPong: total time 161480.833 us, loop 325, bufLen 20510, latency 248.432 u
s. bandwidth 0.082558 GB/s
478680854700:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
478956779572:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
478959703388:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
478960696196:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
478963620012:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
479277916060:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
479280773176:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
479305090974:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
479307948090:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
479313035611:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
479315892727:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
479618132750:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
479620989812:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
479621220960:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
479624078076:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
479669633154:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
479669633154:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
479670216112:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
479670216112:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
479672556788:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
479672556788:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
479673139490:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
479673139490:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
480653761639:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
480653761639:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
480654789486:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
480654789486:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
480656618701:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
480656618701:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
480657646447:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
480657646447:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
AllPingPong: total time 155702.463 us, loop 325, bufLen 9000, latency 239.542 us
. bandwidth 0.037572 GB/s
Allreduce: ranks 32, loop 40, 1 double(s), latency 2511.511 us
Barrier: ranks 32, loop 15, latency 1401.224 us
Simulation is complete, simulated time: 755.876 ms
=== MiniMD dragonfly size 64 (PPN=2) ===
=====
Topology          : Dragonfly (size 64) - MiniMD
hosts_per_router  : 2
routers_per_group : 8
num_groups        : 8
Total routers     : 64
Total nodes       : 128
Total MPI ranks   : 128
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/DragonFly/PPN_2/results/minimd_dragonfly_64_ppn2_nodes128_stats.csv
=====
180000:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
180000:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
180000:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
180000:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128

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[illegible]

[illegible]

180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
161614522406:0:118:EmberEngine:AllPingPong:generate:59: rank=118 size=128
161617446222:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=128
161622780533:0:114:EmberEngine:AllPingPong:generate:59: rank=114 size=128
161625704349:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=128
161761248399:0:112:EmberEngine:AllPingPong:generate:59: rank=112 size=128
161764105461:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=128
161764340611:0:116:EmberEngine:AllPingPong:generate:59: rank=116 size=128
161767197673:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=128
161767428821:0:120:EmberEngine:AllPingPong:generate:59: rank=120 size=128
161770285937:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=128
161772805508:0:109:EmberEngine:AllPingPong:generate:59: rank=109 size=128
161775729324:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128
161798060796:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
161800917912:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=128
161998156127:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
161998650374:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
162000155793:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
162001013189:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=128
162001507463:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=128
162003079609:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128
162004540651:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
162005123609:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
162007464413:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
162007524809:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
162007980287:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
162010381925:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
162016045734:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
162016561992:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
162018902796:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
162019418670:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128
162035777595:0:105:EmberEngine:AllPingPong:generate:59: rank=105 size=128
162036360553:0:97:EmberEngine:AllPingPong:generate:59: rank=97 size=128
162037087583:0:98:EmberEngine:AllPingPong:generate:59: rank=98 size=128
162037537141:0:100:EmberEngine:AllPingPong:generate:59: rank=100 size=128
162038070074:0:99:EmberEngine:AllPingPong:generate:59: rank=99 size=128
162038701229:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=128
162039217231:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=128
162040011217:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=128
162040363220:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
162040460519:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=128
162040772758:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
162040927190:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=128
162042433094:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
162042882396:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
162044029052:0:101:EmberEngine:AllPingPong:generate:59: rank=101 size=128
162044545310:0:103:EmberEngine:AllPingPong:generate:59: rank=103 size=128
162044931503:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
162045343042:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=128
162045343042:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
162045859300:0:102:EmberEngine:AllPingPong:generate:59: rank=102 size=128

162046886114:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=128
162047119930:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=128
162047401988:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=128
162047758249:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
162047855137:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
162048200104:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=128
162048266548:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
162048269171:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
162048274507:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
162048507957:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
162048715978:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=128
162049977046:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
162050339045:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=128
162050615311:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
162050617056:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=128
162051131185:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
162052688713:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
162053289013:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
162053399735:0:104:EmberEngine:AllPingPong:generate:59: rank=104 size=128
162053915993:0:106:EmberEngine:AllPingPong:generate:59: rank=106 size=128
162055497450:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
162055545775:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
162056013708:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
162056061649:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
162056256797:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=128
162056772671:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=128
162057294348:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
162057572487:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
162058354512:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=128
162058870386:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=128
162059416075:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
162059962348:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
AllPingPong: total time 162060.037 us, loop 325, bufLen 20510, latency 249.323 u
s. bandwidth 0.082263 GB/s
162060217982:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
162060495993:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
162060819443:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
162062273191:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
162062819410:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=128
162063676532:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=128
162063710888:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
162064227146:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
162065936667:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
162066567822:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
162067083824:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
162068793783:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
162258917776:0:110:EmberEngine:AllPingPong:generate:59: rank=110 size=128
162261774892:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=128
162263756861:0:108:EmberEngine:AllPingPong:generate:59: rank=108 size=128
162266613977:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=128
162289287620:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=128
162292144736:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
162294361489:0:107:EmberEngine:AllPingPong:generate:59: rank=107 size=128
162297218551:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
162297449699:0:111:EmberEngine:AllPingPong:generate:59: rank=111 size=128
162300306815:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
162310844393:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
162313420347:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128

162313701071:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
162316277463:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
162497346931:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
162498545530:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
162500203993:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=128
162501402363:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=128
162511092467:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=128
162511608725:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=128
162513949529:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
162514465403:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
162532890694:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
162533971901:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
162535728952:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
162536828579:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
162540335081:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
162541541017:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
162541935214:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128
162542621557:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
162543127316:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=128
162544397823:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=128
162544792276:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
162545478774:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
317358836298:0:118:EmberEngine:AllPingPong:generate:59: rank=118 size=128
317360245097:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=128
317368194975:0:114:EmberEngine:AllPingPong:generate:59: rank=114 size=128
317369603774:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=128
317513501592:0:109:EmberEngine:AllPingPong:generate:59: rank=109 size=128
317514910391:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128
317545166083:0:112:EmberEngine:AllPingPong:generate:59: rank=112 size=128
317546641582:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=128
317548258295:0:116:EmberEngine:AllPingPong:generate:59: rank=116 size=128
317549733794:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=128
317551346505:0:120:EmberEngine:AllPingPong:generate:59: rank=120 size=128
317552822004:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=128
317581978480:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
317583453979:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=128
317740851877:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
317742260676:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128
317747608587:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
317749017386:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
317776643764:0:105:EmberEngine:AllPingPong:generate:59: rank=105 size=128
317778052563:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=128
317791384464:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
317792138841:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
317792859963:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
317793614340:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
317800909891:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
317802385390:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
317820553041:0:97:EmberEngine:AllPingPong:generate:59: rank=97 size=128
317821987758:0:99:EmberEngine:AllPingPong:generate:59: rank=99 size=128
317822028540:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=128
317823463257:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=128
317823566547:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
317825042046:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=128
317874093131:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
317875568630:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=128
317911969905:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
317913616156:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128

317938764629:0:101:EmberEngine:AllPingPong:generate:59: rank=101 size=128
317939348537:0:103:EmberEngine:AllPingPong:generate:59: rank=103 size=128
317940410880:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=128
317940994788:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=128
317944095293:0:98:EmberEngine:AllPingPong:generate:59: rank=98 size=128
317944575277:0:100:EmberEngine:AllPingPong:generate:59: rank=100 size=128
317945499995:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=128
317945674844:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=128
317946083903:0:102:EmberEngine:AllPingPong:generate:59: rank=102 size=128
317946192052:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=128
317947022756:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
317947183598:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=128
317947502740:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
317947767506:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=128
317948135312:0:104:EmberEngine:AllPingPong:generate:59: rank=104 size=128
317948256861:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
317948602307:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
317948719220:0:106:EmberEngine:AllPingPong:generate:59: rank=106 size=128
317949011238:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
317949119515:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
317949766282:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
317949781563:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=128
317949940464:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
317950365471:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=128
317950694841:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
317951449885:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
318028237754:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
318028925175:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
318029746252:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=128
318030158398:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
318030809006:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
318031563255:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
318042835460:0:110:EmberEngine:AllPingPong:generate:59: rank=110 size=128
318044310959:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=128
318047674545:0:108:EmberEngine:AllPingPong:generate:59: rank=108 size=128
318049150044:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=128
318065409253:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
318065718613:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
318066472606:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
318067159556:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=128
318067226983:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
318067506140:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=128
318067982027:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
318068326961:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=128
318069081338:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=128
318069836382:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=128
318073205304:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=128
318074680803:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
318075120106:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
318075429466:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
318076183459:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
AllPingPong: total time 156016.653 us, loop 325, bufLen 9000, latency 240.026 us
. bandwidth 0.037496 GB/s
318076870409:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
318077108433:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
318077216993:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
318077692341:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
318078037814:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128

318078279173:0:107:EmberEngine:AllPingPong:generate:59: rank=107 size=128
318078447257:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
318078962788:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
318079546696:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
318079754672:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
318080301612:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
318081367383:0:111:EmberEngine:AllPingPong:generate:59: rank=111 size=128
318082842882:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
318094761410:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
318096236909:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
318097338031:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
318098813530:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
318279403530:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
318281561370:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
318286211599:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
318288369439:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=128
318346711322:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=128
318348186821:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
318392055983:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
318392810360:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
318393531482:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=128
318394285859:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=128
318430063361:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
318431539143:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
318507497008:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=128
318508972507:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
318510173679:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128
318510931903:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
318511649178:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
318512409147:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
318759488941:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
318760964440:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
318766297010:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
318767051387:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
318767772509:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=128
318768526886:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=128
478991011346:0:118:EmberEngine:AllPingPong:generate:59: rank=118 size=128
478993935162:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=128
479014322996:0:114:EmberEngine:AllPingPong:generate:59: rank=114 size=128
479017246812:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=128
479282863866:0:112:EmberEngine:AllPingPong:generate:59: rank=112 size=128
479285720928:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=128
479285956078:0:116:EmberEngine:AllPingPong:generate:59: rank=116 size=128
479288813140:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=128
479289044288:0:120:EmberEngine:AllPingPong:generate:59: rank=120 size=128
479291901404:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=128
479317074963:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
479319932079:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=128
479606124082:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
479608981198:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=128
479621788577:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
479624645693:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=128
479859700140:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
479861992619:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
479862623902:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128
479864916435:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
480025953225:0:109:EmberEngine:AllPingPong:generate:59: rank=109 size=128
480028877041:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128

480032543185:0:105:EmberEngine:AllPingPong:generate:59: rank=105 size=128
480035467001:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=128
480036900696:0:97:EmberEngine:AllPingPong:generate:59: rank=97 size=128
480037416954:0:99:EmberEngine:AllPingPong:generate:59: rank=99 size=128
480039757630:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=128
480040273632:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=128
480176826624:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
480178985703:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
AllPingPong: total time 162102.880 us, loop 325, bufLen 20510, latency 249.389 u
s. bandwidth 0.082241 GB/s
480179750258:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
480181909519:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
480191980197:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
480192666540:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
480194837259:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
480195523757:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
480195762754:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
480198619870:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
480359049690:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
480361906806:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128
480417495565:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
480417734351:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
480419561437:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
480419839448:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
480420878589:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
480423735523:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
480424818558:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
480425334816:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
480427675492:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
480428191494:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
480430145220:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
480431854741:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
480433001898:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
480434711857:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
480525578913:0:101:EmberEngine:AllPingPong:generate:59: rank=101 size=128
480526095171:0:103:EmberEngine:AllPingPong:generate:59: rank=103 size=128
480528435847:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=128
480528951849:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=128
480530081163:0:107:EmberEngine:AllPingPong:generate:59: rank=107 size=128
480531279762:0:111:EmberEngine:AllPingPong:generate:59: rank=111 size=128
480532938225:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
480533346128:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
480533624267:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
480534136595:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
480536269762:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=128
480536547773:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=128
480541425499:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
480541941757:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
480544282561:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=128
480544798435:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=128
480545037304:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
480547894420:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=128
480666465988:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
480666811494:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
480667184347:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
480667633905:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
480669322922:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
480669650246:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=128

480669668300:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
480670108109:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
480670557283:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
480670760134:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
480670978243:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
480671621231:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
480672338200:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
480673617196:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
480673835359:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=128
480674031769:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
480674478448:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
480676607723:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
480676888447:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
480678518678:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=128
480679034936:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=128
480679464839:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
480681375740:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
480681891614:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
480682130483:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=128
480684987599:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
480728751115:0:98:EmberEngine:AllPingPong:generate:59: rank=98 size=128
480728819149:0:100:EmberEngine:AllPingPong:generate:59: rank=100 size=128
480730637564:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=128
480730745618:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=128
480736568355:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=128
480736743109:0:102:EmberEngine:AllPingPong:generate:59: rank=102 size=128
480739425289:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=128
480739600043:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=128
480741037922:0:106:EmberEngine:AllPingPong:generate:59: rank=106 size=128
480741554180:0:104:EmberEngine:AllPingPong:generate:59: rank=104 size=128
480743267703:0:110:EmberEngine:AllPingPong:generate:59: rank=110 size=128
480743894856:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=128
480744410858:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=128
480744977891:0:108:EmberEngine:AllPingPong:generate:59: rank=108 size=128
480746124819:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=128
480747835007:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=128
480853596840:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
480854795439:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
480856453902:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=128
480857652272:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=128
480927339693:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
480930196809:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
481164660961:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128
481165688141:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
481166894077:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
481167517895:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
481167580420:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
481168545075:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
481169751139:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=128
481170437637:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=128
AllPingPong: total time 155749.248 us, loop 325, bufLen 9000, latency 239.614 us
. bandwidth 0.037560 GB/s
Allreduce: ranks 128, loop 40, 1 double(s), latency 2491.712 us
Barrier: ranks 128, loop 15, latency 1405.198 us
Simulation is complete, simulated time: 756.695 ms
All PPN=2 MiniMD dragonfly runs finished. Results in: /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/DragonFly/PPN_2/results
-rw-r--r-- 1 stephenrajva staff 71905 28 Jun 06:59 results/minimd_dragonfly_


```

16_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff   12513 28 Jun 06:59 results/minimd_dragonfly_
4_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  431932 28 Jun 06:59 results/minimd_dragonfly_
64_ppn2_nodes128_stats.csv
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniMD/DragonFly/PPN_2"
cp ../analyze_universal.py .
python3 - << 'EOF'
import re
f = "analyze_universal.py"; src = open(f).read()
src = re.sub(r'^STUDY\s*=.*$', 'STUDY = "MiniMD_Dragonfly_PPN2"', src, cou
nt=1, flags=re.M)
src = re.sub(r'^TOPOLOGY\s*=.*$', 'TOPOLOGY = "other"', src, cou
nt=1, flags=re.M)
runs = '''RUNS = [
    ("4", (2, 2), "minimd_dragonfly_4_ppn2_nodes8_stats.csv"),
    ("16", (4, 4), "minimd_dragonfly_16_ppn2_nodes32_stats.csv"),
    ("64", (8, 8), "minimd_dragonfly_64_ppn2_nodes128_stats.csv"),
]'''
src = re.sub(r'^RUNS\s*=\s*\[.*?\]', runs, src, count=1, flags=re.M | re.S)
src = src.replace('PPN = 1', 'PPN = 2').replace('PPN=1', 'PPN=2')
open(f, "w").write(src)
print("Config updated:")
for line in src.splitlines():
    if line.startswith(("STUDY", "TOPOLOGY", "RUNS")) or 'ppn2_nodes' in line: pri
nt(line)
EOF
python3 analyze_universal.py
Config updated:
STUDY = "MiniMD_Dragonfly_PPN2"
TOPOLOGY = "other"
RUNS = [
    ("4", (2, 2), "minimd_dragonfly_4_ppn2_nodes8_stats.csv"),
    ("16", (4, 4), "minimd_dragonfly_16_ppn2_nodes32_stats.csv"),
    ("64", (8, 8), "minimd_dragonfly_64_ppn2_nodes128_stats.csv"),
]

MiniMD_Dragonfly_PPN2
Size      Latency(ns)  Hops      Thru(GB/s)  Packets    Bits          Idle%    Stalls
-----
4         227.11      -         0.6132      298304     3699969280   83.1    0
16        398.06      -         2.8555     1394232    17267394240  88.0    0
64        561.22      -         12.2249     5978400    74004384000  92.0    72190947
Saved: MiniMD_Dragonfly_PPN2_latency_ns.png
(skipping Average hop count: not available for topology 'other')
Saved: MiniMD_Dragonfly_PPN2_throughput.png
Saved: MiniMD_Dragonfly_PPN2_packets.png
Saved: MiniMD_Dragonfly_PPN2_bits.png
Saved: MiniMD_Dragonfly_PPN2_idle_pct.png
Saved: MiniMD_Dragonfly_PPN2_stalls.png
stephenrajva@Stephens-MacBook-Air-4 PPN_2 %

```